

Backup Live Demo

For

Clickstream Analysis personalization platform

طبقة التشغيل التي تقوم عليها مراحل المشروع

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PROBLEMS OUTPUT TERMINAL
bash - clickstream-personalization-platform + v [ ] [ ] ... | [ ]

waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ source /home/waleed/A1/clickstream-personalization-platform/.venv/bin/activate
(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ docker compose ps -a
NAME                                IMAGE                                COMMAND                                SERVICE    CREATED        STATUS              PORTS
clickstream-airflow                 clickstream-airflow:3.2.2           "/usr/bin/dumb-init ..."          airflow    36 hours ago   Up 16 minutes       0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp
clickstream-clickhouse              clickhouse/clickhouse-server:24.8.14.39 "/entrypoint.sh"                    clickhouse 36 hours ago   Up 22 minutes (healthy) 0.0.0.0:8123->8123/tcp, [::]:8123->8123/tcp, 9009/tcp, 0.0.0.0:9002->9000/tcp, [::]:9002->9000/tcp
clickstream-debezium-connect        debezium/connect:3.0.0.Final         "/docker-entrypoint..."           debezium-connect 36 hours ago   Up 16 minutes       8778/tcp, 0.0.0.0:8083->8083/tcp, [::]:8083->8083/tcp, 9092/tcp
clickstream-filebeat                elastic/filebeat:8.19.17            "/usr/bin/tini -- /u..."          filebeat    36 hours ago   Up 16 minutes
clickstream-kafka-ui               ghcr.io/kafbat/kafka-ui:latest      "/bin/sh -c 'java --..."          kafka-ui    36 hours ago   Up 16 minutes       0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
clickstream-kafka1                  confluentinc/cp-kafka:7.3.2         "/etc/confluent/dock..."          kafka1      36 hours ago   Up 16 minutes (healthy) 9092/tcp, 0.0.0.0:29092->29092/tcp, [::]:29092->29092/tcp
clickstream-kafka2                  confluentinc/cp-kafka:7.3.2         "/etc/confluent/dock..."          kafka2      36 hours ago   Up 22 minutes (healthy) 9092/tcp, 0.0.0.0:29093->29093/tcp, [::]:29093->29093/tcp
clickstream-kafka3                  confluentinc/cp-kafka:7.3.2         "/etc/confluent/dock..."          kafka3      36 hours ago   Up 22 minutes (healthy) 9092/tcp, 0.0.0.0:29094->29094/tcp, [::]:29094->29094/tcp
clickstream-minio                   quay.io/minio/minio:latest          "/usr/bin/docker-ent..."          minio       36 hours ago   Up 22 minutes (healthy) 0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001->9000-9001/tcp
clickstream-observability-ui        clickstream-observability-ui:1.0.0  "streamlit run app.p..."          observability-ui 10 minutes ago Up 10 minutes (healthy) 0.0.0.0:8501->8501/tcp, [::]:8501->8501/tcp
clickstream-postgres               postgres:16                          "docker-entrypoint.s..."          postgres    36 hours ago   Up 22 minutes (healthy) 0.0.0.0:5433->5432/tcp, [::]:5433->5432/tcp
clickstream-spark-engine            clickstream-spark-engine:3.5.0      "tini -g -- bash -lc..."          spark-engine 36 hours ago   Up 16 minutes (healthy) 4040/tcp, 8888/tcp
clickstream-zookeeper              confluentinc/cp-zookeeper:7.3.2     "/etc/confluent/dock..."          zookeeper   36 hours ago   Up 22 minutes       2181/tcp, 2888/tcp, 3888/tcp

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$

```

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PROBLEMS  OUTPUT  TERMINAL  bash - clickstream-per

(.venv) waheed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ python main.py init

Environment file ready.

✓ [#####] STAGE 100% | OVERALL 11% | STEP 1/9 PASSED | Environment check | PASSED | 0s

=====
ENVIRONMENT CHECK [PASSED]
=====

```

STATUS	CHECK	DETAILS
PASS	Python version	3.12.3 (minimum: 3.11)
PASS	docker-compose.yml	Found
PASS	Dockerfile.spark	Found
PASS	Dockerfile.airflow	Found
PASS	requirements.txt	Found
PASS	config/settings.yaml	Found
PASS	config/filebeat.yml	Found
PASS	config/spark-defaults.conf	Found
PASS	.env	Found
PASS	Directory: data/reference	Ready
PASS	Directory: data/source	Ready
PASS	Directory: runtime	Ready
PASS	Directory: reports	Ready
PASS	Directory: data/clickhouse	Ready
PASS	Directory: data/minio/data	Ready
PASS	Spark topology	One spark-engine container is configured
PASS	Kafka topology	3 partitions, RF=3, min ISR=2
PASS	Reference product catalog	Found
PASS	Geolite2 database	Found
PASS	Environment variable: POSTGRES_PASSWORD	Configured
PASS	Environment variable: MINIO_ROOT_PASSWORD	Configured
PASS	Environment variable: CLICKHOUSE_PASSWORD	Configured
PASS	Environment variable: CALENDARIFIC_API_KEY	Configured
PASS	Environment variable: GEOIP_DATABASE_PATH	Configured
PASS	Docker CLI	Docker version 29.4.2, build 055a478
PASS	Docker Compose	Docker Compose version v5.1.3
PASS	Docker daemon memory	10.7 GiB available; low-memory project profile requires at least 10.0 GiB

```

=====
✓ [#####] STAGE 100% | OVERALL 22% | STEP 2/9 PASSED | Source generation | PASSED | 0s

```

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PROBLEMS  OUTPUT  TERMINAL  bash - clickstream-persc

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ python main.py init

```

```

✓ [#####] STAGE 100% | OVERALL 22% | STEP 2/9 PASSED | Source generation | PASSED | 0s

```

```

SOURCE GENERATION [PASSED]

```

STATUS	CHECK	DETAILS
PASS	Static product catalog	reused without modification; 72 products
PASS	Product Catalog	72 static valid products
PASS	Users seed	96 clean seed rows
PASS	Orders seed	48 clean seed rows
PASS	Order items seed	96 clean seed rows
PASS	Mixed Clickstream source	441 lines: 434 valid, 5 invalid, 2 duplicate
PASS	Mixed Web Log source	416 .log lines: 410 valid, 4 invalid, 2 duplicate
PASS	Source layout	Clickstream and Web Logs each use one independent source file

```

✓ [#####] STAGE 100% | OVERALL 33% | STEP 3/9 PASSED | Infrastructure startup | PASSED | 141s

```

```

INFRASTRUCTURE STARTUP [PASSED]

```

STATUS	CHECK	DETAILS
PASS	postgres	Running
PASS	zookeeper	Running
PASS	kafka1	Running
PASS	kafka2	Running
PASS	kafka3	Running
PASS	kafka_ui	Running
PASS	debezium	Running
PASS	minio	Running
PASS	filebeat	Running
PASS	spark	Running
PASS	airflow	Running
PASS	clickhouse	Running
PASS	operations_console	Running
PASS	MinIO	HTTP 200

```
PROBLEMS OUTPUT TERMINAL bash - clickstream-per

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ python main.py init

=====
STATUS  CHECK                DETAILS
=====
PASS    Controlled CDC mutations  updated user=USR000001, updated order=ORD000001, deleted
temporary user=USR999999
=====
✓ [#####] STAGE 100% | OVERALL 89% | STEP 8/9 PASSED | Initial streaming load | PASSED | 255s
=====
STREAMING INITIAL LOAD [PASSED]
=====
STATUS  CHECK                DETAILS
=====
PASS    Initial streaming load  All approved source types completed Spark processing
=====
✓ [#####] STAGE 100% | OVERALL 100% | STEP 9/9 PASSED | Analytics refresh | PASSED | 802s
=====
ANALYTICS REFRESH [PASSED]
=====
STATUS  CHECK                DETAILS
=====
PASS    user_scd2             Spark runner completed user_scd2
PASS    weather_enrichment    Spark runner completed weather_enrichment
PASS    holiday_enrichment     Spark runner completed holiday_enrichment
PASS    validate_lakehouse     Spark runner completed validate_lakehouse
PASS    publish_serving        Spark runner completed publish_serving
=====
Refresh run: initial_refresh_20260708T140107Z

INIT COMPLETE
=====
Platform status: HEALTHY
Streaming remains active. New clickstream, web-log, and PostgreSQL CDC records will continue to be processed.
Open the Operations Console and confirm the latest validation and serving build are PASSED.
=====
(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$
```

kafbat UI

afc9c91 v1.5.0

UTC+3:00

Dashboard

clickstream-cluster

Brokers

Topics

Consumers

clickstream-cluster

Topics

Search by Topic Name

+ Add a Topic

Export CSV

Show Internal Topics

Delete selected topics

Copy selected topic

Purge messages of selected topics

	Topic Name	Partitions	Out of sync replicas	Replication Factor	Number of messages	Size	
☐	IN __consumer_offsets	50	0	3	N/A	25 KB	
☐	clickstream-events	3	0	3	1206	2 MB	
☐	debezium-connect-configs	1	0	3	N/A	27 KB	
☐	debezium-connect-offsets	3	0	3	N/A	3 KB	
☐	debezium-connect-status	3	0	3	N/A	59 KB	
☐	order-items-cdc	3	0	3	96	197 KB	
☐	orders-cdc	3	0	3	49	124 KB	
☐	users-cdc	3	0	3	99	216 KB	
☐	webserver-logs	3	0	3	1181	1 MB	

Debezium CDC Connectors

```
PROBLEMS OUTPUT TERMINAL bash - clickstre

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ python - <<'PY'
import json
import urllib.request
import urllib.parse

BASE = "http://localhost:8083"

def read_json(path):
    with urllib.request.urlopen(BASE + path, timeout=15) as response:
        return json.loads(response.read().decode("utf-8"))

try:
    connectors = read_json("/connectors")
except Exception:
    print("ERROR: Cannot reach Debezium Connect at http://localhost:8083")
    print("Make sure the platform is running.")
    raise SystemExit(1)

print()
print("DEBEZIUM CDC CONNECTORS - STATUS")
print("=" * 95)
print(f"{'Connector Name':45} {'Connector State':18} {'Task States':20}")
print("-" * 95)

for name in connectors:
    encoded_name = urllib.parse.quote(name, safe="")
    status = read_json(f"/connectors/{encoded_name}/status")

    connector_state = status.get("connector", {}).get("state", "?")
    task_states = ", ".join(
        task.get("state", "?") for task in status.get("tasks", [])
    )
    PYint()Expected evidence: users, orders, and order_items CDC connectors are RUNNING.)

DEBEZIUM CDC CONNECTORS - STATUS
=====
Connector Name                Connector State  Task States
-----
clickstream-users-cdc         RUNNING        RUNNING
clickstream-order-items-cdc   RUNNING        RUNNING
clickstream-orders-cdc        RUNNING        RUNNING
-----
Expected evidence: users, orders, and order_items CDC connectors are RUNNING.

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$
```

```
(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ python - <<'PY'
import json
import subprocess

completed = subprocess.run(
    ["python", "main.py", "status"],
    capture_output=True,
    text=True,
    check=True
)

data = json.loads(completed.stdout)

checks = {item["component"]: item for item in data.get("checks", [])}

spark_streaming = checks.get("Spark streaming", {})
validation = checks.get("Latest validation", {})
serving = checks.get("Active serving build", {})

print()
print("SPARK STRUCTURED STREAMING EVIDENCE")
print("=" * 110)
print(f"{'Component':35} {'Status':12} {'Details'}")
print("-" * 110)
print(f"{'Overall Platform Status':35} {'PASS':12} {data.get('overall_status', 'UNKNOWN')}")
print(f"{'Spark streaming':35} {spark_streaming.get('status', 'UNKNOWN'):12} {spark_streaming.get('detail', 'N/A')}")
print(f"{'Latest validation':35} {validation.get('status', 'UNKNOWN'):12} {validation.get('detail', 'N/A')}")
print(f"{'Active serving build':35} {serving.get('status', 'UNKNOWN'):12} {serving.get('detail', 'N/A')}")
print("-" * 110)
print()
PY

SPARK STRUCTURED STREAMING EVIDENCE
=====
Component                               Status    Details
-----
Overall Platform Status                 PASS      HEALTHY
Spark streaming                         PASS      Last completed batch: 236
Latest validation                       PASS      PASSED
Active serving build                    PASS      build_20260711T150653Z__serving
=====

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$
```

Spark Batch Processing

```
(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ python - <<'PY'
from pathlib import Path
import re

log_dir = Path("runtime/airflow_requests/logs") ...
text = path.read_text(errors="ignore")
text_lower = text.lower()
name_lower = path.name.lower()

matched = any(p in name_lower or p in text_lower for p in patterns)
if not matched:
    continue

if "sparkcontext: running spark version" not in text_lower:
    continue

completed = (
    "sparkcontext is stopping with exitcode 0" in text_lower
    or "successfully stopped sparkcontext" in text_lower
)

explicit_failed = (
    '"status": "failed"' in text_lower
    and "sparkcontext is stopping with exitcode 0" not in text_lower
)

if completed and not explicit_failed:
    matches.append((path.stat().st_mtime, path, text))

if not matches:
PYint()-" * 110)_name:28} {app_name[:32]:32} {spark_version:14} COMPLETED")ark"ion Status'}")
```

SPARK BATCH PROCESSING EVIDENCE

Batch Job	Spark Application	Spark Version	Execution Status
User SCD2 Incremental	user-scd2-incremental	Spark 3.5.0	COMPLETED
Weather Enrichment	publish-clickhouse	Spark 3.5.0	COMPLETED
Holiday Enrichment	publish-clickhouse	Spark 3.5.0	COMPLETED
Validate Lakehouse	validate-lakehouse	Spark 3.5.0	COMPLETED
Publish Serving	publish-clickhouse	Spark 3.5.0	COMPLETED

```
(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$
```

PROBLEMS OUTPUT TERMINAL

bash - clickstream-personalization-platform + v [] [] ... |

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● waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ source /home/waleed/A1/clickstream-personalization-platform/.venv/bin/activate
● (.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ docker compose exec airflow bash -lc 'airflow tasks states-for-dag-run analytics_ref
resh "manual__2026-07-13T08:29:31.313137+00:00"'

```

```

2026-07-13T08:51:51.302248Z [info    ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-07-13T08:51:51.302599Z [info    ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-07-13T08:51:51.302821Z [info    ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-07-13T08:51:51.303020Z [info    ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-07-13T08:51:51.303209Z [info    ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-07-13T08:51:51.303483Z [info    ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37

```

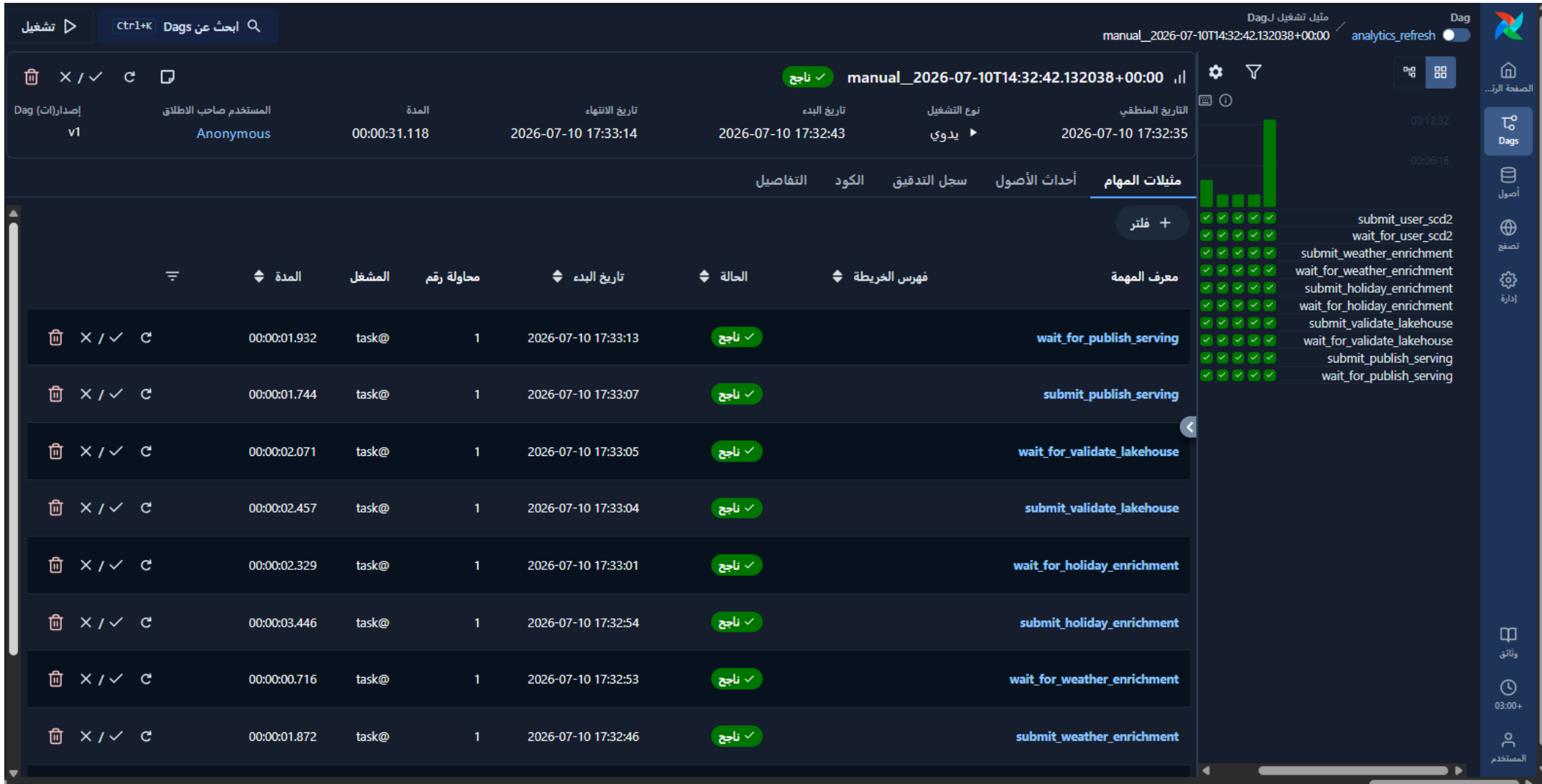
dag_id	logical_date	task_id	state	start_date	end_date
analytics_refresh	2026-07-13T08:29:29+00:00	submit_user_scd2	success	2026-07-13T08:35:35.802450+00:00	2026-07-13T08:35:37.056605+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	wait_for_user_scd2	success	2026-07-13T08:35:42.254733+00:00	2026-07-13T08:35:43.096971+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	submit_weather_enrichment	success	2026-07-13T08:35:44.022637+00:00	2026-07-13T08:35:45.119810+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	wait_for_weather_enrichment	success	2026-07-13T08:35:46.282289+00:00	2026-07-13T08:35:46.774434+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	submit_holiday_enrichment	success	2026-07-13T08:35:51.876540+00:00	2026-07-13T08:35:52.862086+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	wait_for_holiday_enrichment	success	2026-07-13T08:35:58.120160+00:00	2026-07-13T08:35:58.733524+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	submit_validate_lakehouse	success	2026-07-13T08:36:03.849840+00:00	2026-07-13T08:36:04.612233+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	wait_for_validate_lakehouse	success	2026-07-13T08:39:07.166045+00:00	2026-07-13T08:39:07.685363+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	submit_publish_serving	success	2026-07-13T08:39:12.787292+00:00	2026-07-13T08:39:13.139802+00:00
analytics_refresh	2026-07-13T08:29:29+00:00	wait_for_publish_serving	success	2026-07-13T08:39:14.112996+00:00	2026-07-13T08:39:14.634432+00:00

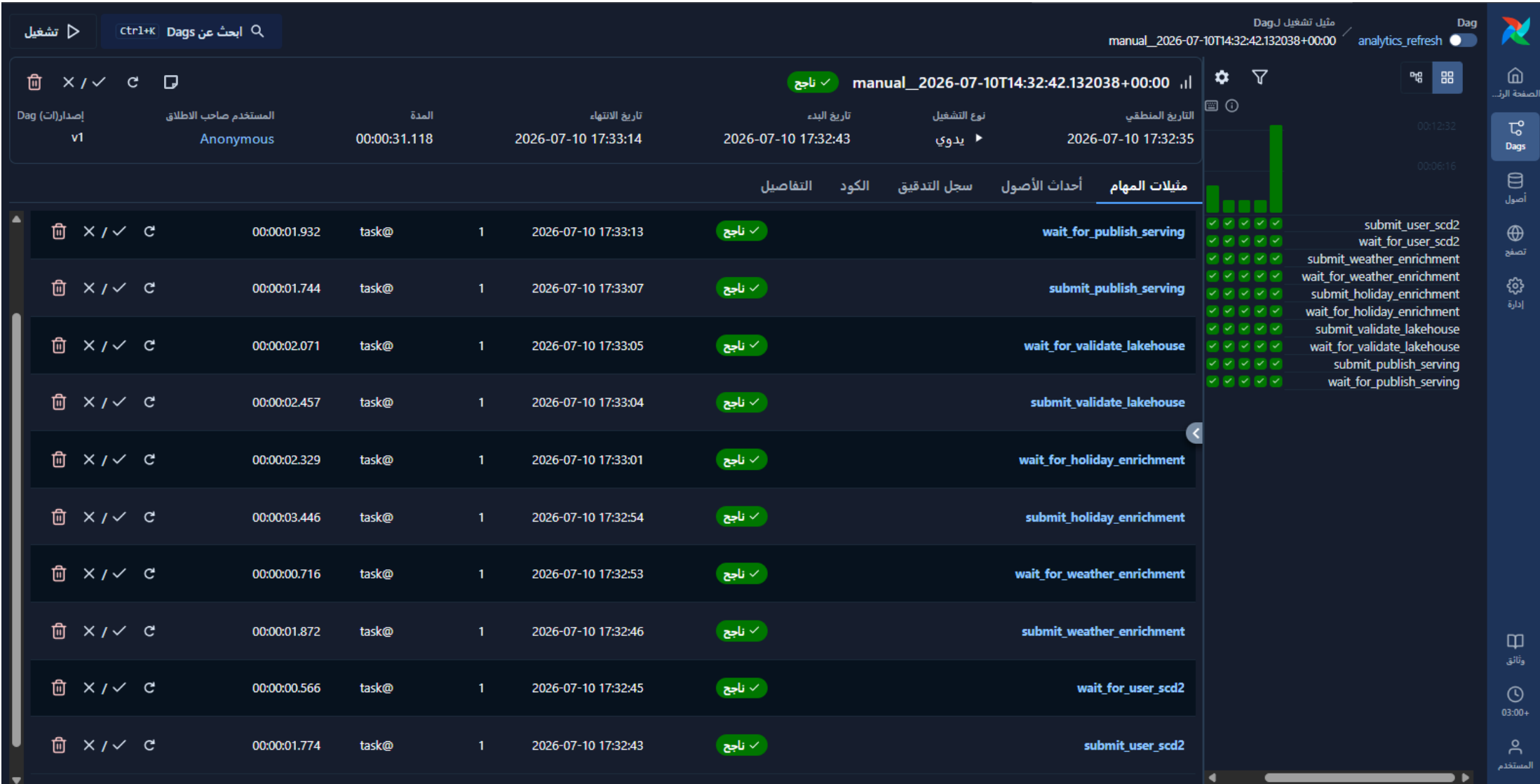
```

○ (.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$

```

Airflow DAG





```

PROBLEMS  OUTPUT  TERMINAL
bash - clickstream-personalization

(.venv) waheed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$ LOG_FILE=$(
for f in $(ls -t runtime/airflow_requests/logs/*validate_lakehouse*.log 2>/dev/null); do
    grep -q "ecommerce.raw.kafka_messages" "$f" &&
    grep -q "ecommerce.processed.clickstream_clean" "$f" &&
    grep -q "ecommerce.processed.webserver_logs_clean" "$f" && ...
printf -- "%.0s-" {1..120}; printf "\n"

for item in \
    "Raw Kafka Messages|ecommerce.raw.kafka_messages" \
    "Clickstream Clean|ecommerce.processed.clickstream_clean" \
    "Webserver Logs Clean|ecommerce.processed.webserver_logs_clean" \
    "Quarantine Records|ecommerce.audit.quarantine_records"
do
    LABEL="${item%%|*}"
    TABLE="${item##*|}"
    EVIDENCE=""

    grep -q "Table loaded by catalog: $TABLE" "$LOG_FILE" && EVIDENCE="${EVIDENCE}LOADED, "
    grep -q "Scanning table $TABLE" "$LOG_FILE" && EVIDENCE="${EVIDENCE}SCANNED, "
    grep -q "Pushing operators to $TABLE" "$LOG_FILE" && EVIDENCE="${EVIDENCE}QUERY_PUSHED, "

    EVIDENCE="${EVIDENCE%, }"
    [ -z "$EVIDENCE" ] && EVIDENCE="FOUND"

    printf "%-30s %-45s %-30s\n" "$LABEL" "$TABLE" "$EVIDENCE"
done
else
    printf "No complete lakehouse validation evidence was found.\n"
fi

printf -- "%.0s-" {1..120}; printf "\n\n"

SPARK LAKEHOUSE READ EVIDENCE
=====
Evidence Log: scheduled_validate_lakehouse.log
-----


| Lakehouse Table      | Iceberg Table Identifier                 | Spark Evidence                |
|----------------------|------------------------------------------|-------------------------------|
| Raw Kafka Messages   | ecommerce.raw.kafka_messages             | LOADED, SCANNED, QUERY_PUSHED |
| Clickstream Clean    | ecommerce.processed.clickstream_clean    | LOADED, SCANNED, QUERY_PUSHED |
| Webserver Logs Clean | ecommerce.processed.webserver_logs_clean | LOADED, SCANNED, QUERY_PUSHED |
| Quarantine Records   | ecommerce.audit.quarantine_records       | LOADED, SCANNED, QUERY_PUSHED |


-----
(.venv) waheed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform$

```

MINIO
OBJECT STORE
Community Edition

[←](#)

[+](#) Create Bucket

Buckets

[ecommerce-lakehou...](#)

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Object Browser



ecommerce-lakehouse

Created on: **Thu, Jul 09 2026 05:20:36 (GMT+3)** Access: **PRIVATE** 225.1 MiB - 5337 Objects

Rewind

Refresh



ecommerce-lakehouse / warehouse / raw / kafka_messages / data / ingested_at_day=2026-07-09



▲ Name

Last Modified



source_name=clickstream



source_name=order_items_cdc



source_name=orders_cdc



source_name=users_cdc



source_name=web_logs

MINIO

OBJECT STORE

Community Edition

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 Create Bucket

Q

 Filter Buckets

Buckets

ecommerce-lakehou...

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Object Browser

Q

 Start typing to filter objects in the bucket

ecommerce-lakehouse

Created on: Thu, Jul 09 2026 05:20:36 (GMT+3) Access: PRIVATE 219.8 MiB - 5268 Objects

Rewind

Refresh

<

ecommerce-lakehouse / warehouse / audit

Crea

☐	▲ Name	Last Modified
☐	external_api_failures	
☐	pipeline_runs	
☐	quality_metrics	
☐	quarantine_records	
☐	serving_builds	
☐	validation_runs	
☐	watermarks	

MINIO

OBJECT STORE

Community Edition

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Create Bucket

Q

Filter Buckets

Buckets

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Q

Start typing to filter objects in the bucket

ecommerce-lakehouse

Created on: **Thu, Jul 09 2026 05:20:36 (GMT+3)** Access: **PRIVATE** 219.8 MiB - 5268 Objects

Rewind

Refresh

<

ecommerce-lakehouse / warehouse / processed

Cre

ClickHouse Serving Views عدد الصفوف في

PROBLEMS

OUTPUT

TERMINAL

waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform\$ source /home/waleed/A1/clickstream-personalization-platform/.venv/bin/activate

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform\$ docker compose exec -T clickhouse bash -lc 'clickhouse-client \

```
--user "$CLICKHOUSE_USER" \
--password "$CLICKHOUSE_PASSWORD" \
--database personalization_olap \
--query "
SELECT '\v_dim_product\' AS object_name, count() AS rows FROM v_dim_product
UNION ALL
SELECT '\v_dim_user_current\'', count() FROM v_dim_user_current
UNION ALL
SELECT '\v_fact_clickstream_event\'', count() FROM v_fact_clickstream_event
UNION ALL
SELECT '\v_fact_order\'', count() FROM v_fact_order
UNION ALL
SELECT '\v_fact_order_item\'', count() FROM v_fact_order_item
UNION ALL
SELECT '\v_mart_journey_session\'', count() FROM v_mart_journey_session
UNION ALL
SELECT '\v_mart_navigation_paths\'', count() FROM v_mart_navigation_paths
UNION ALL
SELECT '\v_mart_context_impact_daily\'', count() FROM v_mart_context_impact_daily
FORMAT PrettyCompact
"
```

object_name

rows

1. v_fact_clickstream_event 1284

object_name

rows

2. v_mart_navigation_paths 96

object_name

rows

3. v_mart_context_impact_daily 54

object_name

rows

4. v_fact_order_item 96

object_name

rows

5. v_dim_product 72

object_name

rows

6. v fact order 48

bash - clickstream-personalization-platform

(.venv) waleed@DESKTOP-KMUS1B4:~/A1/clickstream-personalization-platform\$ docker compose exec -T clickhouse bash -lc 'clickhouse-client \

```
--user "$CLICKHOUSE_USER" \
--password "$CLICKHOUSE_PASSWORD" \
...
UNION ALL
SELECT '\v_fact_clickstream_event\'', count() FROM v_fact_clickstream_event
UNION ALL
SELECT '\v_fact_order\'', count() FROM v_fact_order
UNION ALL
SELECT '\v_fact_order_item\'', count() FROM v_fact_order_item
UNION ALL
SELECT '\v_mart_journey_session\'', count() FROM v_mart_journey_session
UNION ALL
SELECT '\v_mart_navigation_paths\'', count() FROM v_mart_navigation_paths
UNION ALL
SELECT '\v_mart_context_impact_daily\'', count() FROM v_mart_context_impact_daily
FORMAT PrettyCompact
"
```

object_name

rows

1. v_dim_product 72

object_name

rows

2. v_fact_order_item 96

object_name

rows

3. v_mart_journey_session 948

object_name

rows

4. v_fact_clickstream_event 1284

object_name

rows

5. v_fact_order 48

object_name

rows

6. v_dim_user_current 97

object_name

rows

7. v_mart_context_impact_daily 54

object_name

rows

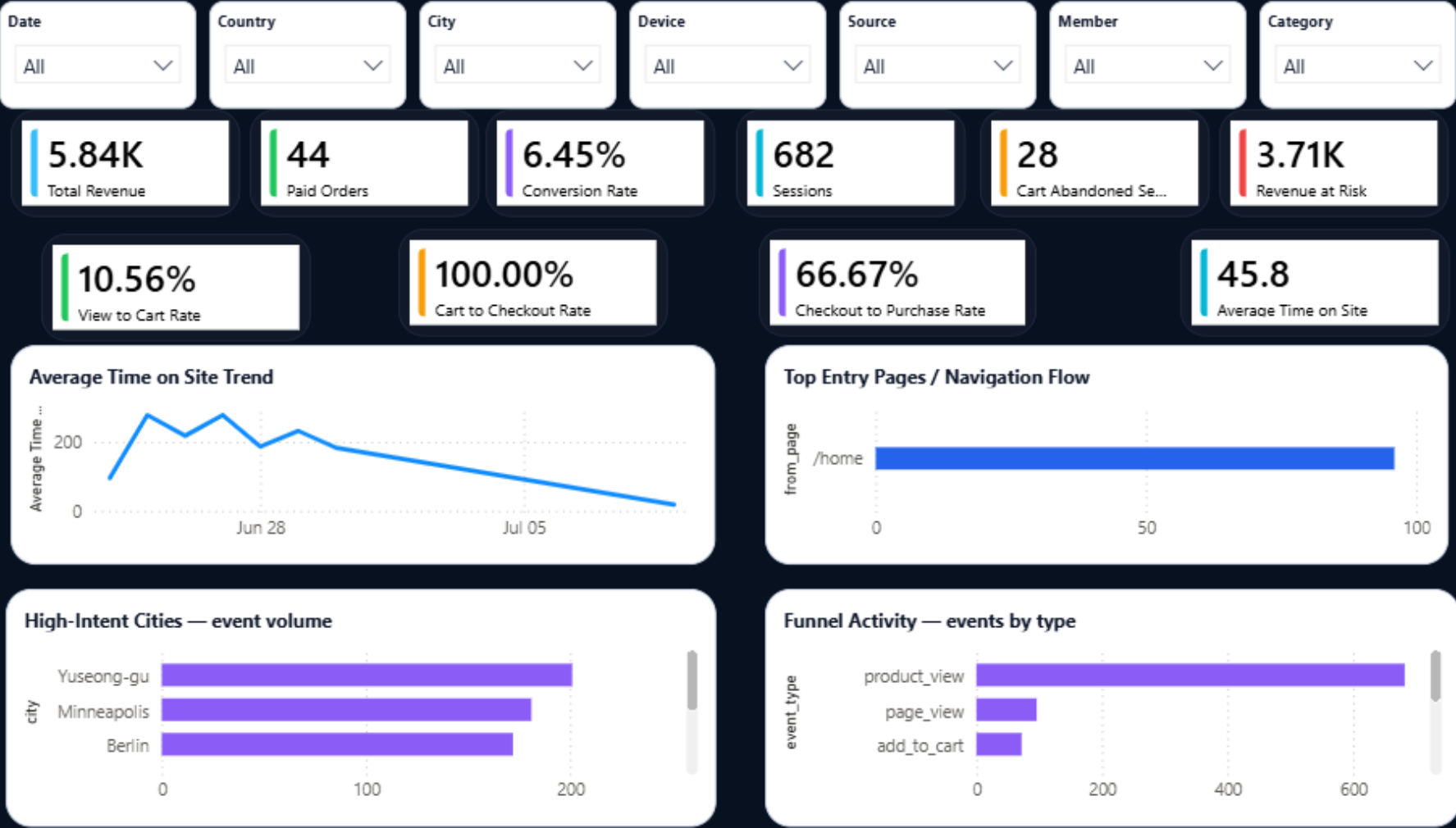
8. v_mart_navigation_paths 96

E-Commerce
Decision Intelligence

- Growth + Funnel
- Personalization

Growth, Funnel Leakage & Journey Intelligence

Revenue, conversion leakage, engagement, navigation flow, and high-intent demand signals.



Power BI Serving Layer

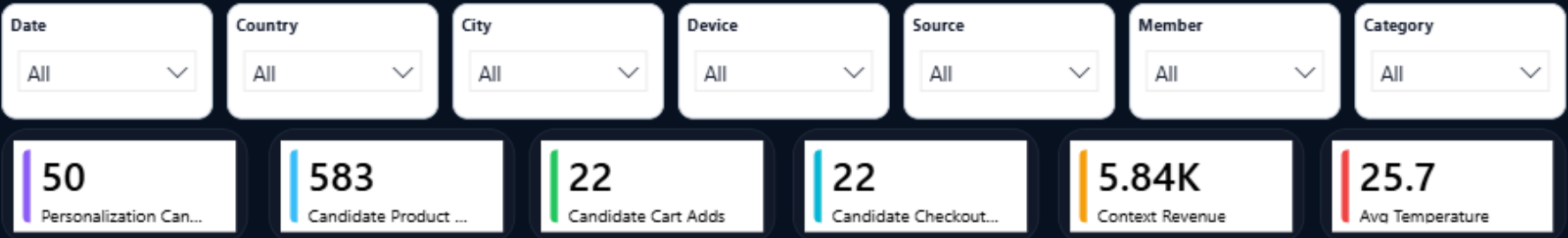
E-Commerce
Decision
Intelligence

Growth + Funnel

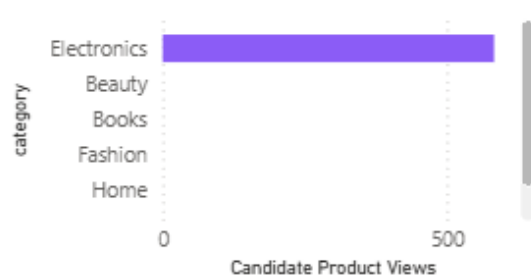
Personalization

Personalization, Context & Recommendation Intelligence

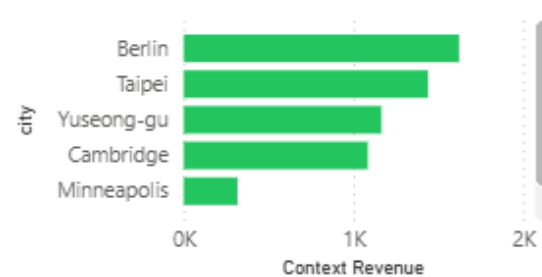
Next-best products, segment demand, geography, membership, holiday, and weather context signals.



Recommendation Demand by Category



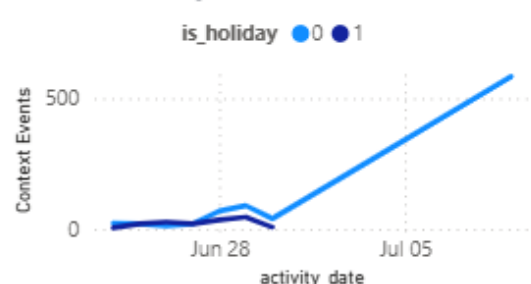
Context Revenue by City



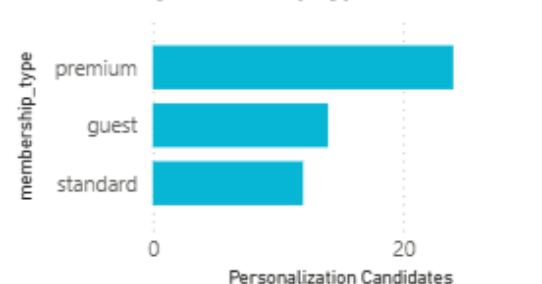
Next-Best Product Candidates

membership_type	city	product_name	
guest	Taipei	Cotton Hoodie 09	F
guest	Taipei	Gaming Mouse 25	E
guest	Taipei	Leadership Journal 66	E
guest	Taipei	Perfume Mist 34	E
guest	Taipei	Portable Charger 31	E
guest	Taipei	Storage Basket 20	F
guest	Taipei	Training Gloves 29	S
guest	Yuseong-gu	Cotton Hoodie 09	F
guest	Yuseong-gu	Gaming Mouse 25	E
guest	Yuseong-gu	Leadership Journal 66	E
guest	Yuseong-gu	Perfume Mist 34	E
guest	Yuseong-gu	Portable Charger 31	E
guest	Yuseong-gu	Storage Basket 20	F
guest	Yuseong-gu	Training Gloves 29	S
premium	Berlin	Body Lotion 22	E
premium	Berlin	Coffee Maker 02	F
premium	Berlin	Face Cleanser 52	E
premium	Berlin	Hair Dryer 10	E
Total			

Observed Holiday Association — events



Candidates by Membership Type



PROBLEMS	OUTPUT	TERMINAL						docker - clickstream-pe
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS	
b3dcbf86bcd	clickstream-airflow	8.19%	670.3MiB / 700MiB	95.76%	23.1kB / 22.4kB	323MB / 641MB	30	
cb9023814339	clickstream-spark-engine	278.57%	2.496GiB / 2.5GiB	99.82%	65MB / 17.6MB	829MB / 500MB	677	
6c1de40bef80	clickstream-debezium-connect	3.40%	441.7MiB / 450MiB	98.16%	2.09MB / 1.86MB	351MB / 32.4MB	70	
853532e33902	clickstream-filebeat	0.04%	80.81MiB / 128MiB	63.13%	25.6kB / 78.3kB	195MB / 1.04MB	11	
9d8c7bbe09e8	clickstream-kafka-ui	0.04%	225.8MiB / 240MiB	94.09%	873kB / 172kB	157MB / 45.1MB	28	
2a94c6f04954	clickstream-kafka1	3.24%	288.9MiB / 384MiB	75.24%	1.87MB / 1.98MB	22.6MB / 74.9MB	84	
a9f3ca7ce6a2	clickstream-kafka3	2.88%	286.2MiB / 384MiB	74.53%	1.87MB / 2.7MB	35.2MB / 85.7MB	88	
ccf925d32652	clickstream-kafka2	3.35%	280.6MiB / 384MiB	73.08%	2.06MB / 2.04MB	18.5MB / 80.6MB	84	
1cdad5f61191	clickstream-observability-ui	0.00%	57.58MiB / 320MiB	17.99%	2.81kB / 126B	30MB / 791kB	4	
09f410f3302c	clickstream-postgres	8.63%	96.68MiB / 450MiB	21.49%	758kB / 1.22MB	48MB / 57.6MB	18	
e565b7feee84	clickstream-clickhouse	7.37%	424.2MiB / 1GiB	41.43%	48.9kB / 50.8kB	348MB / 46MB	678	
2e3c95e06241	clickstream-zookeeper	0.11%	119.2MiB / 320MiB	37.24%	187kB / 178kB	59.3MB / 1.42MB	46	
49ee051f6e2a	clickstream-minio	22.73%	86.23MiB / 256MiB	33.68%	17.2MB / 63.8MB	119MB / 17.6MB	12	
870fb1e196f6	portainer	0.00%	18.24MiB / 10.7GiB	0.17%	1.53kB / 126B	67MB / 524kB	7	